

MATH I - Week 2

The following material helps clarify concepts, methods, and examples (presented in Lecture) - it also presents answers to students' questions.

1. Bacterial Growth ¶

Bacteria reproduce mainly through *binary fission*: one bacterial cell divides into two, the two become four, then eight, and so forth. Under ideal conditions—sufficient nutrients, suitable temperature and no limiting factors—the population therefore grows approximately exponentially.

If N_0 is the initial number of bacteria and the population doubles every T_d hours, then after time t ,

$$N(t) = N_0 2^{t/T_d}.$$

The equivalent exponential form is

$$N(t) = N_0 e^{kt},$$

where $k > 0$ is the growth-rate constant.

This is due to the fact that $2^{t/T_d} = e^{kt}$, with the relationship between k and the doubling time T_d being

$$k = \frac{\ln 2}{T_d}, \quad T_d = \frac{\ln 2}{k}.$$

Example

A bacterial culture initially contains $N_0 = 500$ cells and doubles every 30 minutes. Its population after t hours is

$$N(t) = 500 2^{t/0.5} = 500 2^{2t}.$$

After two hours,

$$N(2) = 500 2^{2/0.5} = 500 2^4 = 500(16) = 8\,000.$$

Thus, the culture grows from 500 to 8 000 bacteria in two hours.

Time (hours)	Number of doublings	Population
0	0	500
0.5	1	1 000
1.0	2	2 000
1.5	3	4 000
2.0	4	8 000

The continuous growth-rate constant is

$$k = \frac{\ln 2}{0.5} \approx 1.386 \text{ h}^{-1},$$

so the same model can be written as

$$N(t) = 500e^{1.386t}.$$

NOTE: exponential growth is a good model for the early growth phase. The bacterial population cannot grow exponentially forever because limited resources.

2. The Richter Scale and Earthquake Magnitudes

The Richter scale is a logarithmic scale used to describe the magnitude of an earthquake—that is, the amount of seismic energy released at its source. It was introduced in 1935 by Charles F. Richter for earthquakes recorded in Southern California. Today, seismologists commonly use the more accurate moment magnitude scale, M_w , particularly for large earthquakes. Nevertheless, earthquake magnitudes are still frequently described informally as “Richter-scale” values.

In a simplified model, earthquake magnitude can be written as

$$M = \log_{10} \left(\frac{A}{A_0} \right),$$

where A is the measured amplitude of the seismic waves and A_0 is a reference amplitude. Because the logarithm has base 10, an increase of one magnitude unit corresponds to a tenfold increase in measured wave amplitude:

$$\frac{A_2}{A_1} = 10^{M_2 - M_1}.$$

Thus, a magnitude 6 earthquake produces seismic waves with approximately ten times the amplitude of those produced by a magnitude 5 earthquake. A magnitude 7 earthquake produces approximately $10^2 = 100$ times the wave amplitude of a magnitude 5 earthquake.

Earthquake energy increases even more rapidly. An increase of one magnitude unit corresponds to approximately 31.6 times more released energy. The energy ratio of two earthquakes can be estimated by

$$\frac{E_2}{E_1} \approx 10^{1.5(M_2 - M_1)}.$$

Therefore, **magnitude is not a linear measurement**: a magnitude 8 earthquake is not simply “twice as strong” as a magnitude 4 earthquake.

Example 1: Comparing magnitudes 5 and 7

Consider two earthquakes with magnitudes

$$M_1 = 5, \quad M_2 = 7.$$

The magnitude difference is

$$\Delta M = 7 - 5 = 2.$$

The ratio of their seismic-wave amplitudes is

$$\frac{A_2}{A_1} = 10^2 = 100.$$

The magnitude 7 earthquake therefore produces waves with approximately 100 times greater amplitude.

The energy ratio is

$$\frac{E_2}{E_1} \approx 10^{1.5(2)} = 10^3 = 1\,000.$$

Thus, the magnitude 7 earthquake releases approximately 1,000 times more energy than the magnitude 5 earthquake.

Example 2: Comparing magnitudes 5.2 and 6.5

Suppose one earthquake has magnitude 5.2 and another has magnitude 6.5. The difference is

$$\Delta M = 6.5 - 5.2 = 1.3.$$

The amplitude ratio is

$$\frac{A_2}{A_1} = 10^{1.3} \approx 19.95.$$

Therefore, the magnitude 6.5 earthquake produces waves with approximately 20 times greater amplitude.

The energy ratio is

$$\frac{E_2}{E_1} \approx 10^{1.5(1.3)} = 10^{1.95} \approx 89.1.$$

It releases approximately 89 times more energy than the magnitude 5.2 earthquake.

As a general guide, earthquakes below magnitude 3 are often too weak to be noticed, while earthquakes of magnitude 5 can cause local damage. Magnitudes 6 and 7 can produce serious or major regional damage, and earthquakes of magnitude 8 or greater can be devastating. However, the actual damage also depends on depth, distance from populated areas, local geology, building design and construction quality. Magnitude measures the earthquake at its source: it does not by

3. Inverse functions

their usefulness, how to define/determine them, and examples of systems for which Inverse Functions are important

Usefulness of *Inverse Functions* in engineering: they are useful when a system has a known *input–output relationship*, but we need to reverse the relationship: from the output to the input.

If

$$y = f(x),$$

then the inverse function gives

$$x = f^{-1}(y).$$

Here are several practical examples.

1. Temperature conversion

Celsius temperature C is converted to Fahrenheit F by

$$F(C) = \frac{9}{5}C + 32.$$

To recover Celsius from a Fahrenheit measurement, use the inverse:

$$C(F) = F^{-1}(F) = \frac{5}{9}(F - 32).$$

For example, if $F = 95^\circ\text{F}$,

$$C = \frac{5}{9}(95 - 32) = 35^\circ\text{C}.$$

2. Sensor calibration

Suppose a temperature sensor produces voltage

$$V(T) = 0.02T + 1,$$

where T is measured in degrees Celsius and V in volts. The sensor measures voltage, but the engineer needs temperature. Solving for T ,

$$T(V) = V^{-1}(V) = \frac{V - 1}{0.02}.$$

If the measured voltage is 2.2 V, then

$$T = \frac{2.2 - 1}{0.02} = 60^\circ\text{C}.$$

Thus, calibration often consists of finding or approximating an inverse function.

3. Electrical resistance thermometer

A simple resistance-temperature model is

$$R(T) = R_0(1 + \alpha T).$$

The instrument measures R , while temperature is obtained from

$$T(R) = \frac{R/R_0 - 1}{\alpha}.$$

For

$$R_0 = 100 \, \Omega, \quad \alpha = 0.004 \, ^\circ\text{C}^{-1},$$

a measured resistance of $120 \, \Omega$ corresponds to

$$T = \frac{120/100 - 1}{0.004} = 50^\circ\text{C}.$$

4. Sound intensity and decibels

Sound level is related to intensity by

$$L(I) = 10 \log_{10} \left(\frac{I}{I_0} \right),$$

where I_0 is a reference intensity. The inverse relationship is

$$I(L) = I_0 10^{L/10}.$$

If $L = 80$ dB, then

$$I = I_0 10^8.$$

Thus, an 80-dB sound has an intensity 10^8 times the reference intensity.

5. Radioactive dating

The remaining amount of a radioactive material is

$$N(t) = N_0 e^{-\lambda t}.$$

If N is measured, the age is found using the inverse:

$$t(N) = -\frac{1}{\lambda} \ln \left(\frac{N}{N_0} \right).$$

If half the original material remains, then

$$t = \frac{\ln 2}{\lambda},$$

which is the material's half-life.

NOTE: Inverse functions appear naturally in calibration, measurement, system identification, imaging,

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